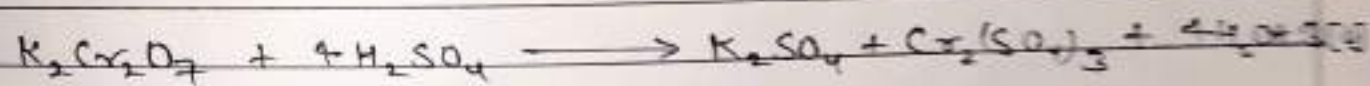


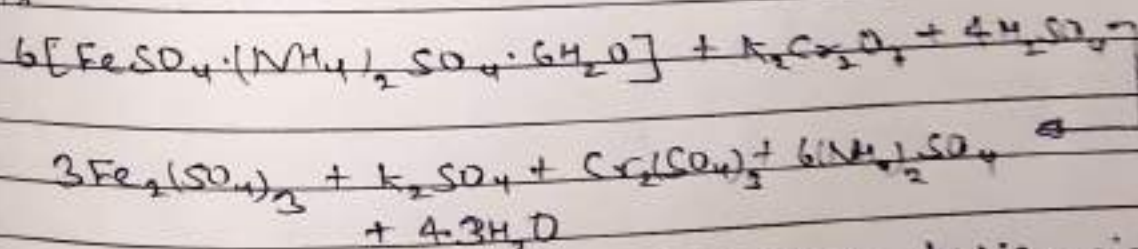
Object :- Determine the strength of a Unknown ferrous ammonium sulphate (FAS) solution by titrating it with Pot. Dichromate solution using potassium ferricyanide as an external indicator.

Apparatus / Reagents required :- Burette (50ml), pipette (10ml), Conical flask (100ml), FAS solution, known solution of $K_2Cr_2O_7$, potassium ferricyanide indicator, 10% Sulphuric Acid.

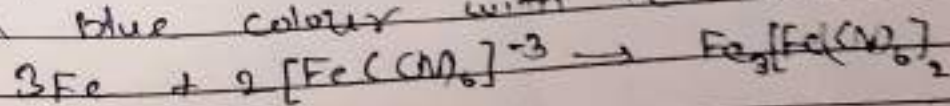
Theory :- Potassium dichromate is good oxidizing reagent and in presence of dilute H_2SO_4 it seems forms chromic sulphate as shown by following equation.



The nascent oxygen so liberated oxidized Fe^{2+} ion of FAS into ferric Fe^{3+} ion. The main reaction is



The external indicator potassium ferricyanide give greenish blue colour with Fe^{2+} ion.



Procedure:-

1. Prepare standard solution FAS of given strength by dissolving the required amount of FAS in distilled water in 100 ml volumetric flask.
2. Fill the burette with known $K_2Cr_2O_7$ solution.
3. Take 10 ml of standard (FAS) solution & 10 ml Sulphuric acid in a conical flask.
4. Start adding same amount of $K_2Cr_2O_7$ from burette to conical flask.
5. Take a white tile & arrange few drops of external indicator on it.
6. Take a drop of analyte from flask & mixed with a drop of external indicator on white tile.
7. If blue colour appears then it means end point is not come yet, add more amount of $K_2Cr_2O_7$ from burette.
8. Repeat the procedure to check the analyst after addition of every drops of $K_2Cr_2O_7$.
9. If no blue coloration is observed on white tile then it shows that end point comes.
10. Repeat the titration till we get three concordant reading.
11. Follow the titration procedure for the Unknown solution to obtain the burette reading.

Calculation:-

For standard FAS solution with $K_2Cr_2O_7$

$$N_1 V_1 = N_2 V_2$$

$$N_1 \times 11 = N_2 \times 10$$

30

$$N_1 = \frac{N_2}{11 \times 3}$$

here

N_1 = Normality of $K_2Cr_2O_7$

N_2 = Normality of known FAS = N_{30}

V_1 = Volume of $K_2Cr_2O_7$

V_2 = 10 ml

Now for unknown FAS solution with same $K_2Cr_2O_7$

$$N_3 V_3 = N_4 V_4$$

$$N_3 = N_4$$

$$\frac{N}{11 \times 3} \times 6.3 = N_4 \times 10$$

$$N_4 = \frac{6.3}{11 \times 3} N$$

$$N_4 = 0.0209 N$$

Strength of unknown FAS solution

$$= \text{Normality of FAS} \times \text{eq weight of FAS}$$

$$= 0.0209 \times 39216$$

$$= 8.196 \text{ gm/l}$$

Observation:-

Titration between standard FAS solution & intermediate $K_2Cr_2O_7$

S. No	Volume of FAS Solution (ml)	Volume of $K_2Cr_2O_7$		Difference (ml)	Concordant reading (ml)
		Initial (ml)	Final (ml)		
1.	10 ml	0.0	11.0	11.0	11.0
2.	10 ml	11.0	22.0	11.0	

Titration between unknown FAS solution and $K_2Cr_2O_7$

S. No	Volume of FAS Solution (ml)	Volume of $K_2Cr_2O_7$		Difference (ml)	Concordant reading (ml)
		Initial (ml)	Final (ml)		
1.	10 ml	0.0	6.3	6.3	6.3
2.	10 ml	6.3	12.6	6.3	

Calculation:-

Strength of unknown FAS solution:

$$= \text{Normality of FAS} \times \text{eq weight of FAS}$$

$$= 8.196 \text{ gm/l}$$

Result: The strength of unknown FAS solution is 8.196 gm/lit

Precautions:-

- (i) All the apparatus should be washed properly.
- (ii) $K_2Cr_2O_7$ solution should be added drop wise.
- (iii) End point should be observed carefully.

Industrial Application:-

- (a) It is used to determine the strength of unknown solution.
- (b) Titration is used to analyse acid rain.
- (c) Titrations are used to determine the concentration in surface water.
- (d) It is also used in food & beverage industry to analyse the standard conc. of specific additives.
- (e) It is used in wine industry where SO_2 is used to control microbial growth. Too much / too little however, can be detrimental to the equality of final product. Titration enabled wineries to determine the conc. of SO_2 during the process.
- (f) FAS is used in the Frick's dose meter to measure high doses of gamma rays.

Viva Questions

Q1 Why dil. H_2SO_4 is added in the preparation of FAS solution?

Ans Dil. H_2SO_4 is essential in FAS preparation to maintain the integrity & effectiveness of the ferrous ammonium sulfate solution.

Q2 What do you mean by Volumetric analysis?

Ans Volumetric analysis is a quantitative chemical analysis method used to determine the concentration of an unknown solution by measuring its volume against a solution of known concentration (standard solution). It is commonly performed through titration.

Q3 What do you understand by concordant reading?

Ans A concordant reading refers to a set of consistent and closely agreeing values obtained in a titration experiment. These reading help ensure accuracy & reliability in volumetric analysis.

Q4 Explain the type of indicator?

Ans Indicator are substance that show a visible change in color or physical property to signal the completion of a reaction, particularly in titration experiment.

It has different types -

(A) Acid-Base indicator - are used for pH dependent titration.

ex. Phenolphthalein, Methyl orange.

(B) Redox indicator - are used in oxidation-reduction reaction.

ex. KMnO_4 , Starch.

(C) Complexometric indicator - are used in metal ion titration.

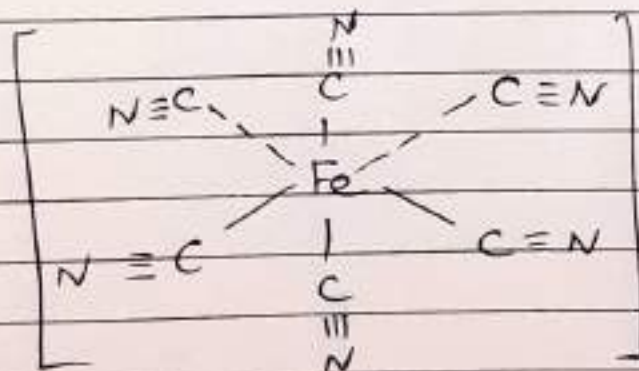
ex. EBT

(D) Precipitation indicator - helps detect precipitate formation.

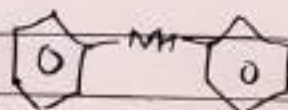
ex. fluorescein

Q5 Which indicator is used in this titration? write its structure.

Ans Potassium ferricyanide, external indicator is used in this titration.



dyp
diphy.
diphenylamine
 $\text{C}_6\text{H}_5-\text{NH}-\text{C}_6\text{H}_5$



Q6 Explain the principle involved in this titration.
 Ans It is based on an oxidation-reduction reaction between Fe^{2+} (ferrous ion) in FAS solution & MnO_4^- (permanganate ion) in acidic medium, or $\text{Cr}_2\text{O}_7^{2-}$ (dichromate ion). An external indicator like diphenylamine is required, which change color at end point of titration (colorless to blue).

Q7 How will you prepare 1 lit 0.1 N $\text{K}_2\text{Cr}_2\text{O}_7$ solution.

Ans Formula used -

$$\text{weight (g)} = \text{Normality} \times \text{eq. weight} \times \text{Volume} \\ = 0.1 \times \frac{294.2}{6} \times 1 = 4.903 \text{ g is used}$$

steps -> weigh 4.903 g of $\text{K}_2\text{Cr}_2\text{O}_7$, dissolve in 500 ml of distilled water in a 1L volumetric flask.
 -> Add dil H_2SO_4 , make up to 1000 ml with distilled water.

Mix well & store in bottle.

Q8 Why distilled water is used to prepare std solution?

Ans Because it doesn't contain impurities like dissolved ions.